M.W.P. DEWMI ABEYKOON

Data Science Undergraduate

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Portfolio | LinkedIn | GitHub

SUMMARY

A passionate final-year Data Science undergraduate with a strong foundation in Mathematics and a keen interest in Artificial Intelligence, Machine Learning, and Data Analytics. Enjoys learning new technologies, conducting research, and developing practical solutions through hands-on projects. Curious, adaptable, and committed to continuous learning, with a strong analytical mindset and problem-solving approach. Eager to apply academic knowledge to real-world challenges while growing both personally and professionally in the field of Data Science.

TECHNICAL SKILLS

- Programming Languages: Python, SQL, Java, JavaScript, Kotlin
- Data Science & ML: Machine Learning, Deep Learning, Scikit-learn, TensorFlow, NLP, Predictive Modeling
- Big Data: Apache Spark , PySpark
- BI & Data Visualization: Power BI, Streamlit
- Databases & Cloud: MongoDB, Firebase
- Development & Platforms: Android Studio, Docker, Git/GitHub

ACADEMIC & PERSONAL PROJECTS

DeepShield — Explainable Multimodal Framework for Fake News Detection **Ongoing**

Final-Year Research Project | Python, PyTorch, CLIP, BERT, RAG, FAISS, FastAPI, React.js

- Designing a multimodal fake news detection framework that fuses pretrained image (CLIP) and text (BERT) models with metadata analysis and RAG-based evidence verification to assess news credibility.
- Applying Explainable AI techniques (Grad-CAM, SHAP) to generate an interpretable Trust Score and human-readable justifications, moving beyond traditional binary classification.
- Building a web-based prototype (FastAPI + React.js) evaluated on Fakeddit, FakeNewsNet, and LIAR datasets using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Co-authoring research under SLTC Research University, Department of Software Engineering.

KAMATHA — AI-Powered Plant Disease Detection App

TensorFlow, CNN, JavaScript, Gemini API

- Contributed to the development of an AI-powered mobile application enabling real-time plant disease detection from leaf imagery using a CNN model.
- Integrated the Gemini API to deliver AI-generated diagnostics and natural/chemical treatment recommendations, alongside an interactive chatbot assistant.
- Implemented multi-language support and persistent chat history to enhance usability across diverse user bases.
- *Live site: kamathaweb.netlify.app, Live App: kamathaapp.netlify.app*

SmartCare AI — Appointment No-Show Prediction System

Python, Scikit-Learn, XGBoost, SHAP, Streamlit

- Built an AI-powered system to predict hospital appointment no-shows using machine learning.
- Trained and compared 5 classification models with SHAP-based explainability.
- Developed an interactive Streamlit dashboard for real-time predictions and analytics.
- *Live Demo : smartcare-ai-project-g11.streamlit.app*

Customer Churn Prediction — End-to-End MLOps Pipeline

Python, Machine Learning, MLOps, MLflow,

- Architected and deployed an end-to-end machine learning pipeline for customer churn prediction, spanning data preprocessing, feature engineering, model training, and deployment.
- Applied MLOps best practices, including basic Docker containerization, to ensure model reproducibility, version control, and streamlined retraining workflows.

Supermarket Business Intelligence — Data Analytics Case Study

Python, SQL, Power BI

- Executed end-to-end analysis of a large-scale supermarket sales dataset, including data cleaning, preprocessing, and statistical hypothesis testing.
- Trained and evaluated machine learning models to extract actionable business insights from sales and customer behavior trends.
- Engineered an interactive Power BI dashboard translating analytical findings into decision-ready visualizations for stakeholders.

LankaSmartMart — Android Shopping Application

Kotlin, Android Studio, Firebase

- Engineered a full-featured Android e-commerce application with secure user authentication and real-time data synchronization via Firebase.
- Designed a scalable app architecture supporting product browsing, cart management, and end-to-end purchase workflows.

Student Management System

Java

- Developed a Java-based application implementing structured CRUD logic for efficient management of student records.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Data Science — Undergraduate (Final Year) **2023 – 2027**

SLTC Research University

G.C.E. Advanced Level — Physical Science Stream **2022**

Visakha Girls' High School, Badulla

G.C.E. Ordinary Level **2018**

Visakha Girls' High School, Badulla

CERTIFICATIONS & ACHIEVEMENTS

- Introductory SQL — Kaggle (June 2026)
- Advanced SQL — Kaggle (June 2026)
- Certificate of Comprehensive UI/UX Design — FSD Academy (August 2025); completed 7 projects spanning mobile app, full website, and client-based UX process flow work
- Certificate of Participation, "Will AI Replace Your Business or Transform It?" Webinar — Sri Lanka Technology Campus (June 2026)
- Certificate of Appreciation, Codemania v5.0 (12-Hour Algorithmic Coding Competition), Team Pixel Mind — IEEE Computer Society Student Branch Chapter, SLTC (February 2025)

SOFT SKILLS & AWARDS

- Leadership
- Communication
- Problem-Solving
- Critical Thinking
- Team Collaboration
- Problem Solving
- Award: Certificate of Appreciation, Codemania v5.0 Algorithmic Coding Competition (2025)

LANGUAGES

English | Sinhala

REFERENCES

References available upon request.